

GUOQING ZHANG

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PROFESSIONAL SUMMARY

Robotics controls and systems engineer with a Ph.D. in Mechanical Engineering and 5+ years developing and validating continuum/medical robotic systems across kinematic and dynamic modeling, motion-control analysis, state estimation, sensor fusion, hardware/software integration, and experimental test infrastructure. Experienced with C++/Python/MATLAB, ROS2, Simulink Real-Time/Speedgoat, EtherCAT, SolidWorks, and complex electromechanical prototypes. First-author IEEE Transactions on Robotics paper and accepted IEEE/ASME Transactions on Mechatronics paper; industry R&D experience with Johnson & Johnson and Auris Health.

TECHNICAL SKILLS

Programming & Tools: C++, Python, MATLAB/Simulink, Git, Linux, ROS2/ROS

Robotics & Controls: motion-control analysis, kinematics, statics/dynamics, system identification, EKF/IMM

Real-Time & Verification: Simulink Real-Time, Speedgoat, EtherCAT, 1 kHz logging, algorithm validation

Hardware & Sensing: cable-driven/continuum robots, motor/servo drives, embedded I/O, EM/IMU/vision/tension/torque sensing, SolidWorks

EXPERIENCE

Robotics & Controls Consultant — Johnson & Johnson (Ottava Program), Remote *Oct 2021 – May 2022*

- Derived continuum-instrument kinematics, differential kinematics, and geometric Jacobians for a cable-driven surgical wrist with leader insertion/roll; supported prototype analysis, motion-fidelity evaluation, and design iteration.
- Implemented a configuration-space EKF observer estimating $(\delta, \theta, i, \phi)$ from OptiTrack pose; derived analytic measurement Jacobians and added a quaternion-based orientation formulation to reduce rotation redundancy.
- Built a pose-error analysis pipeline using Gaussian position ellipsoids and Bingham/quaternion orientation statistics, showing typical errors below 5 mm / 5° and identifying noise-driven worst cases; proposed online compensation using recursive/Kalman-style gain estimation for backlash/extension effects.

System Engineer Intern — Auris Health (Johnson & Johnson), Remote *May 2020 – Aug 2020*

- Built a Simulink Real-Time (Speedgoat) rapid-prototyping workflow with auto C code generation, enabling online tuning and 1 kHz cycle-level logging; shortened control-algorithm iteration from days to hours.
- Integrated real-time I/O across Elmo drives, EM tracker, sheath/IDM hardware, and torque sensing; debugged a distributed electromechanical platform for closed-loop system verification.
- Delivered validated demos for IDM motor characterization using sine/chirp excitation and System Identification Toolbox transfer-function fits, plus instrument-control features including tensioning/init and quintic trajectory planning.

Graduate Researcher — Stevens Institute of Technology, Hoboken, NJ *Aug 2019 – May 2026*

- Developed reduced-order stochastic shape estimation in polynomial-curvature state space; implemented adaptive multi-model filtering (IMM-EKF) for robust estimation under sparse/noisy sensing and varying deformation regimes.
- Designed experimental validation pipelines for cable-driven continuum robots using camera/ArUco pose feedback, proximal cable-tension sensing, and distal load-cell references; converted raw sensor logs into estimation and force-inference metrics.
- Built an integrated shape-force estimation framework using polynomial-curvature kinematics and virtual work; achieved 0.54 mm arclength-averaged shape-position MAE and 0.468 N overall force MAE with the PCK2 model.
- Developed SE(3)-based shape-wrench estimation simulations using sparse IMU orientations, Lie-group-consistent EKF updates, rank/conditioning analysis, and runtime-characterized solver comparisons.

EDUCATION

Ph.D., Mechanical Engineering (GPA: 3.97/4.0) — Stevens Institute of Technology, Hoboken, NJ *Aug 2019 – May 2026*

M.S., Mechanical Engineering (GPA: 3.7/4.0) — Columbia University, New York, NY *Jul 2017 – Dec 2018*

SELECTED PUBLICATIONS

- **G. Zhang** and L. Wang, “Stochastic Adaptive Estimation in Polynomial Curvature Shape State Space for Continuum Robots,” **IEEE Transactions on Robotics**, 2026, doi: 10.1109/TRO.2025.3637147.
- **G. Zhang**, Z. Chen, and L. Wang, “Integrated Shape-Force Estimation for Continuum Robots: A Virtual-Work and Polynomial-Curvature Framework,” **IEEE/ASME Transactions on Mechatronics**, accepted, 2026.